Startup and Innovation

WasteWise AI – Intelligent Waste Segregation as a Service”

WasteWise AI is a low-cost, AI-powered smart waste-segregation system that removes the need for users to decide which bin to use. A single input accepts the waste, AI identifies the waste type, and an automated mechanical mechanism routes it to the appropriate compartment.

“Throw once. WasteWise does the rest.”

- Four target streams: Wet • Dry • Recyclable • E-waste
- Initial deployment focus: colleges, offices and residential societies



The Problem

Waste segregation usually depends on correct human behaviour at the point of disposal. In real environments, users frequently put waste in the wrong compartment, causing recyclable material to become contaminated and increasing manual sorting effort.

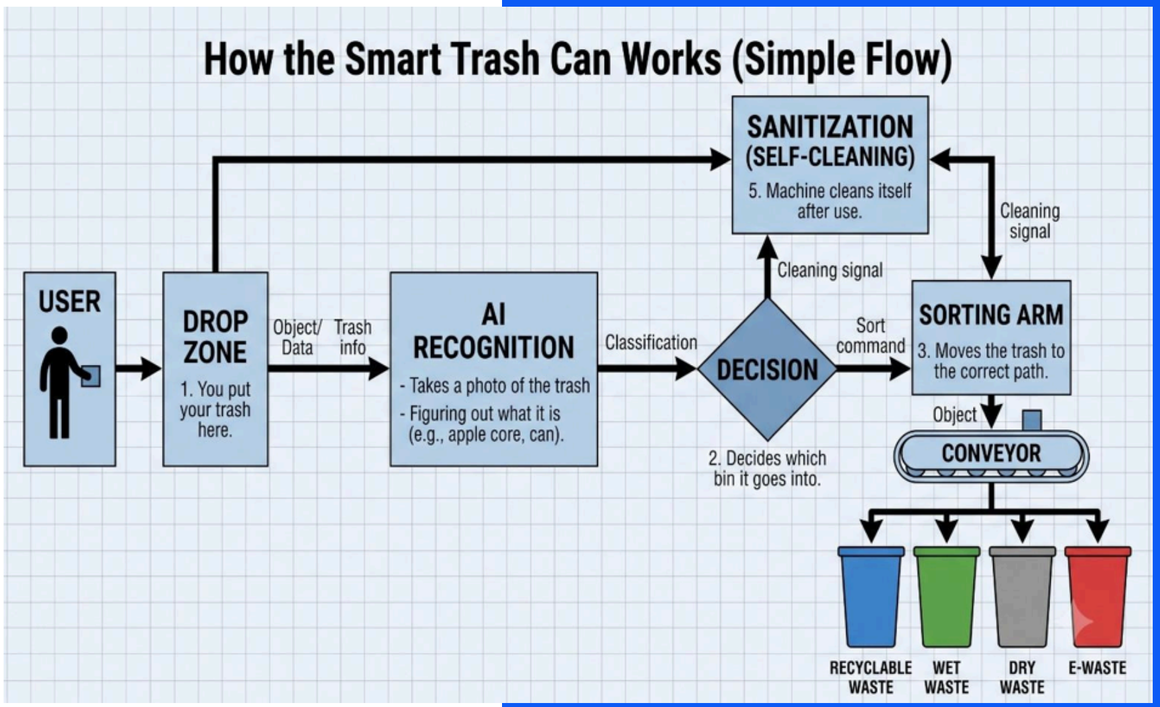
- Mixed waste reduces recycling efficiency.
- Users may be unsure about the correct category.
- Manual segregation is time-consuming and unhygienic.
- Institutions have limited visibility into waste patterns.
- The opportunity is to automate segregation at the source.



Our Solution

WasteWise AI converts a conventional multi-bin station into an intelligent automatic sorting point. The user only needs to throw the item into the common input chamber.

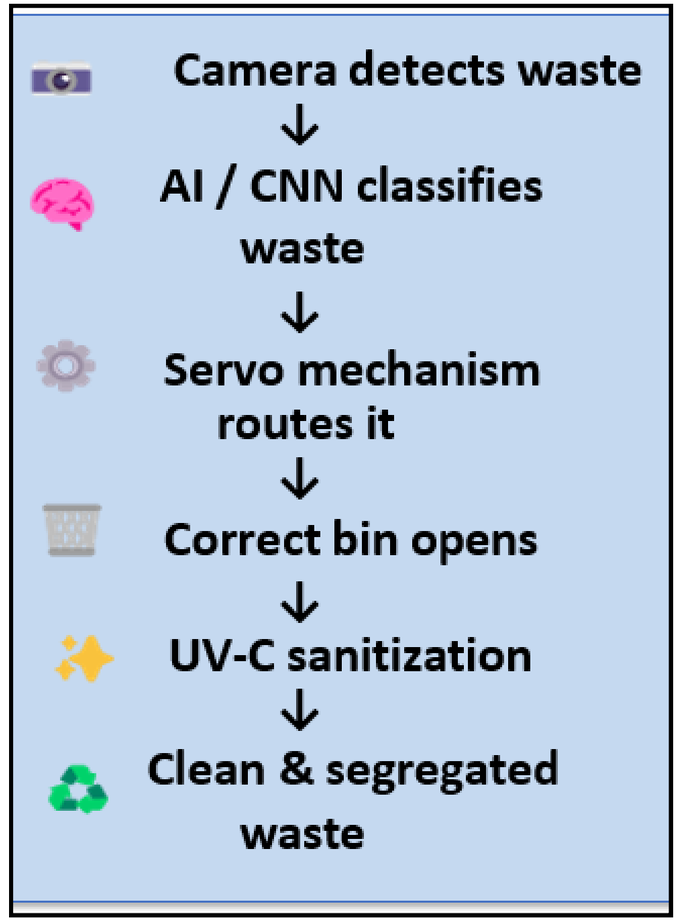
- Sensor detects that an item has entered.
- Camera captures the waste image.
- AI classifies the item.
- Controller selects the destination.
- Servo/mechanical routing system moves the item through the selected path.
- The item falls into the correct internal compartment.



Product & Technology

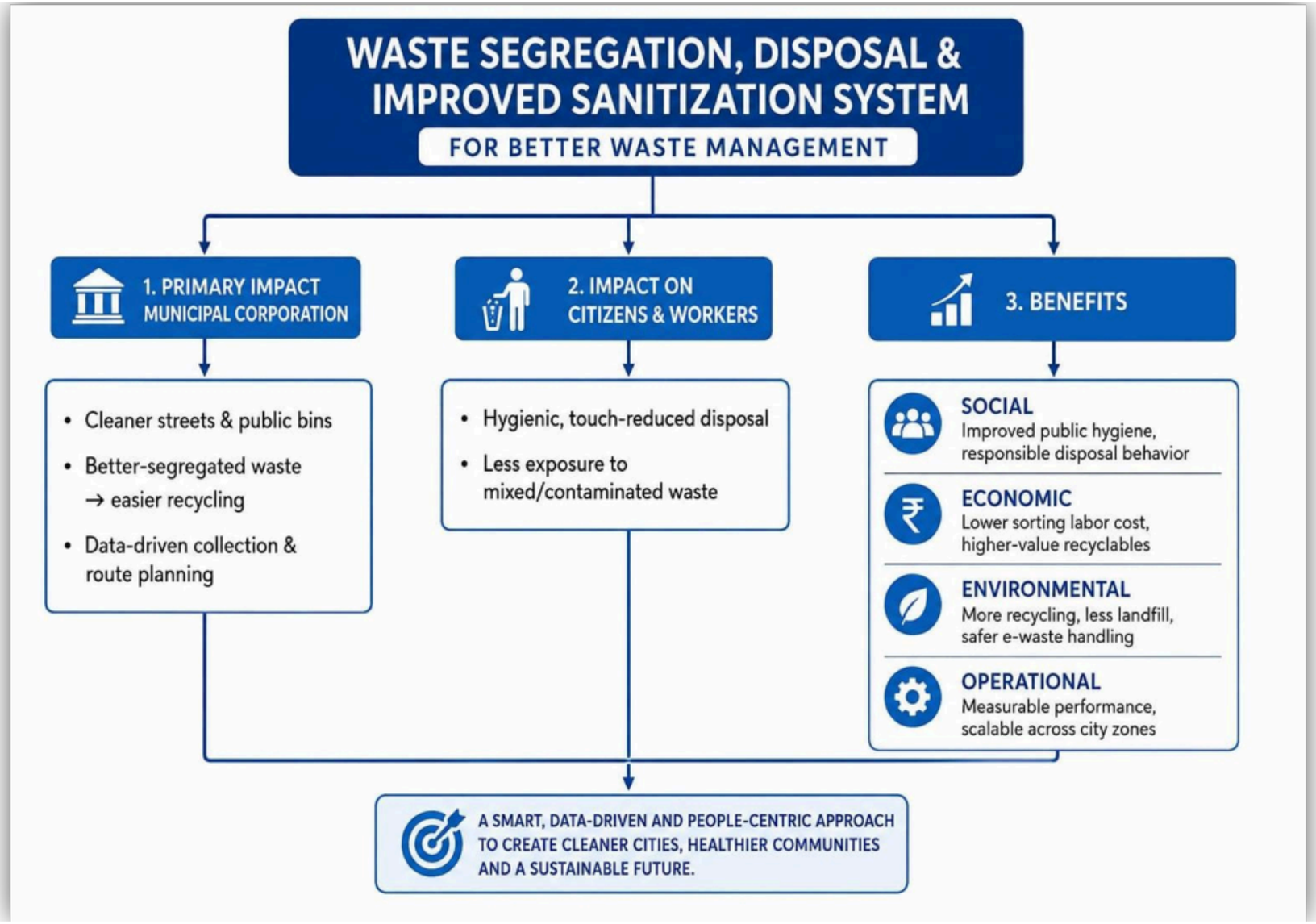
The prototype combines inexpensive electronics, computer vision and a simple mechanical routing mechanism. The design is intentionally modular so the AI and electronics can be upgraded without rebuilding the entire enclosure.

- Camera: ESP32-CAM or USB camera depending on prototype setup.
- Controller: Arduino Nano/ESP32 for hardware control.
- Actuation: servo motors.
- Detection: camera plus entry/position sensor.
- AI: lightweight waste-classification model running locally on a laptop/edge device for the prototype.
- Optional monitoring: bin-level sensor and dashboard in the next version.



Target Customers & Market Entry

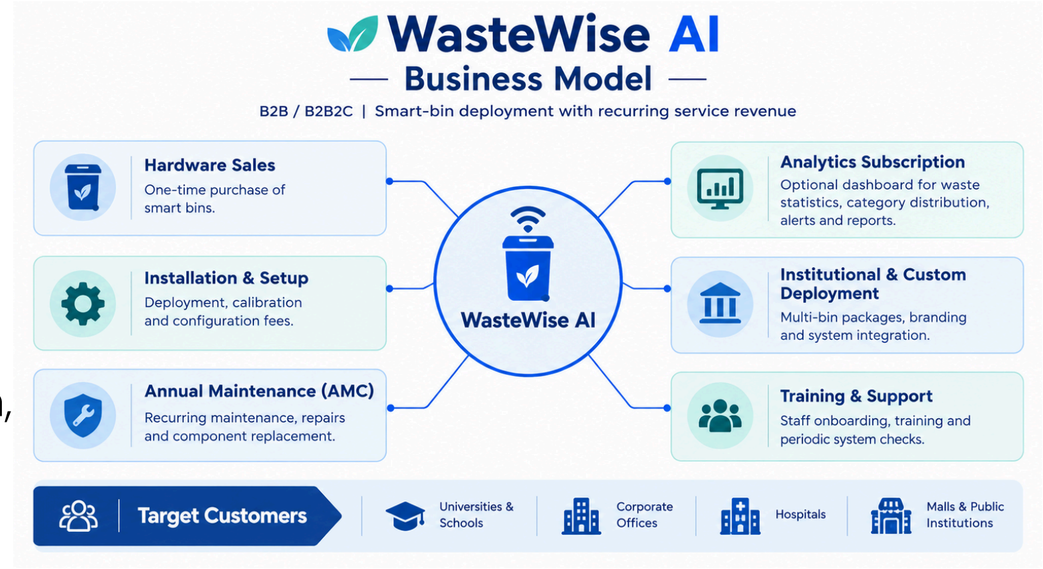
The first customers should be controlled environments where waste volume is predictable and the organization can measure the impact of a pilot.



Business Model

WasteWise follows a B2B/B2B2C model. Institutions purchase or lease the smart bins, while recurring revenue can come from maintenance, analytics and replacement services

- **Revenue Stream 1 — Hardware sale**
  one-time purchase of each smart bin.
- **Revenue Stream 2 — Installation & setup**
  deployment, calibration and configuration fee.
- **Revenue Stream 3 — AMC**
  annual maintenance, repairs and component replacement.
- **Revenue Stream 4 — Analytics subscription**
  optional dashboard for waste statistics, category distribution, alerts and
  reports.
- **Revenue Stream 5 — Institutional/custom deployment**
  multiple-bin packages, branding and integration.
- **Revenue Stream 6 — Training/service**
  staff onboarding and periodic system checks.



Pricing Strategy

The prototype is designed around a low-cost bill of materials, while the commercial product is priced for reliability, enclosure quality, support and software value rather than simply adding a markup to components.

- Basic commercial **WasteWise unit**: approximately **₹6,999.**
- Smart monitored unit with **bin-level sensing/dashboard connectivity**: approximately **₹9,999.**
- **Campus pilot package**: 5 units at approximately **₹44,995** (₹8,999/unit average) including basic setup.
- **Annual AMC**: approximately **₹1,200–₹1,800 per unit/year** depending on service level.
- **Optional analytics**: **₹299–₹499 per unit/month** for institutions that want reporting and alerts.

These are proposed student-startup pricing assumptions for the pitch and should be validated with suppliers and pilot customers before commercialization.

Unit Economics & Revenue Example

A simple commercial example shows how the business can earn from both the initial sale and recurring services.

- Illustrative unit economics: estimated production cost **₹4,000** per monitored unit.
- Illustrative selling price: **₹9,999**.
- Gross contribution before overheads: about **₹5,999** per unit.
- If 100 units are sold at **₹9,999**, gross sales are approximately ₹9.99 lakh.
- If 60% of customers take a **₹1,500** annual AMC, recurring first-year AMC revenue from 100 units would be approximately **₹90,000.**
- At scale, component purchasing, manufacturing and enclosure standardization can reduce per-unit cost.
- Important: these figures are projections, not current market-validated financial results.

Competitive Advantage, Roadmap & Feasibility

WasteWise is differentiated by combining automatic physical segregation with AI classification in a low-cost form factor. The project can start as a campus prototype and grow only after measured validation.

- **Phase 1**: working 4-category prototype.
- **Phase 2**: improve classification dataset and mechanical reliability.
- **Phase 3**: campus pilot with 1–5 units.
- **Phase 4**: add fill-level monitoring and analytics.
- **Phase 5**: commercial manufacturing and institutional contracts.

Key KPIs: classification accuracy, correct-routing rate, sorting time, failure rate, maintenance frequency and waste-category statistics.

Meet Our Team

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Thank You